DEWDRIVE · USER REQUIREMENTS

DewDrive user requirements

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DOCUMENT CONTROL

Document ID	DWD-URS-001	Version	0.1
Status	Draft	Date	2026-10-25
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/dewdrive		
Source	docs/03a-user-requirements.md @ 4f86e9e		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First issue (kit 1.8 pipeline), drawn from the users and stakeholders in DWD-PRB-001 v0.8

DewDrive user requirements

These are the needs of the people DewDrive is for, written from their side. They come from the users and stakeholders in the problem statement (DWD-PRB-001, Tables 1 and 2) and are proposals until co-design sessions with the first partner confirm them. The product requirements that answer them are in DWD-PRS-001 (`03-requirements.md`), and the links are in the generated traceability matrix DWD-RTM-001 (`03b-traceability.md`). Method: STANDARDS section 22.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Performance	Daily water in the dry season	I get a useful daily supplement of drinking water from the air on a typical dry-season night (about 40 % RH)	High	Rural household; school or clinic (PRB Table 1)
UR-002	Performance	Water on the driest nights	I still get some water on the driest nights (about 25 % RH)	Medium	Rural household (PRB Table 1)
UR-003	Usability	Little daily effort	Running it takes two short actions a day that anyone in the household or school can do	High	Rural household; school or clinic
UR-004	Safety	Water I can make safe	The water it makes contains nothing that simple boiling or disinfection cannot make safe to drink	High	Rural household; water and sanitation NGO (PRB Table 2)
UR-005	Safety	No harm to people	Nobody, including children, can be burned, hurt by chemicals or get an electric shock in normal use	High	School or clinic; rural household
UR-006	Environment and durability	Survives an unattended site	It survives sun, dust, wind, rain and animals at a site visited only twice a day	High	Rural household; field station (PRB Table 2)
UR-007	Deployment and assembly	Two people can move and set it up	Two people can carry it in parts and set it up with a spanner and a screwdriver	Medium	Local technician; rural household
UR-008	Logistics	No fuel, no grid	It runs on sunlight alone, with no fuel and no grid connection	High	Rural household (PRB constraints)
UR-009	Maintenance	Built and repaired locally	A local workshop can build and repair it from parts sold by general suppliers, with common tools	High	Local technician or maker; local workshops (PRB Table 2)
UR-010	Cost	Worth the money	The parts cost is low enough that the water is worth it compared with carrying or buying water	High	Rural household; NGO
UR-011	Performance	Comparable yield records	Each unit records its own yield and weather so that sites can be compared	Medium	Researcher or NGO; field station
UR-012	Safety	Salt stays out of the water	The drying salt never gets into the drinking water or spills onto people	High	Rural household; water and sanitation NGO

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.

> **Safety:** UR-004, UR-005 and UR-012 exist because of hazards: untreated water, hot surfaces inside the box, a corrosive salt solution and a small lithium battery. They cannot be waived without a documented decision by Amish.